Lecture 28

Reconstruction and the boundary of quantum groups

Objectives. *The group, its coefficient algebra and its tensor representations encode the same global information in different forms. Differentiation connects these descriptions to the enveloping algebra. It loses information, however, unless the allowed group representations are retained. We make that connection explicit and identify the structures that a later course on quantum groups will deform.*

By the end of this lecture you will be able to:

- Construct the pairing between $U(\mathfrak{k}_{\mathbb{C}})$ and representative functions, with the correct multiplication order.
- For connected K, explain why representative functions form a restricted part of an algebraic dual.
- Recover compact and complex groups from the appropriate tensor automorphisms.
- Use the circle and the $\mathrm{SU}(2)/\mathrm{SO}(3)$ pair to identify what global data survive differentiation.

28.1 The two algebras

Let K be a compact Lie group and $\mathfrak{k}$ its real Lie algebra. As in Lecture 14, $\mathcal{R}(K)$ is the span of the coefficients

$$c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)v), \qquad \lambda \in V^*,\ v \in V,$$

of finite-dimensional continuous complex representations. Every sum of coefficients is one coefficient of a direct sum. Its Hopf structure is

$$\Delta_{\mathcal{R}} f(g,h) = f(gh), \qquad \varepsilon_{\mathcal{R}} f = f(e), \qquad S_{\mathcal{R}} f(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

These formulas and their targets were proved in Lecture 14. We use the ordinary algebraic tensor product. Thus $\Delta_{\mathcal{R}} f$ is a finite sum of tensors, not an infinite expansion.

On the other side, put $U = U(\mathfrak{k}_{\mathbb{C}})$, the universal enveloping algebra from Lie Algebras. It is generated by $\mathfrak{k}_{\mathbb{C}}$ with $XY - YX = [X,Y]$. Its standard Hopf maps are determined by

$$\Delta_U X = X \otimes 1 + 1 \otimes X, \qquad \varepsilon_U X = 0, \qquad S_U X = -X. \tag{28.1}$$

Here Δ_U and ε_U extend as algebra homomorphisms. The antipode S_U extends so that it reverses products. This extension is well defined because reversing the defining relation

sends $XY - YX - [X,Y]$ to $YX - XY + [X,Y]$, which is again zero in U. We check that Δ_U respects the defining relation:

$$[\Delta_U X, \Delta_U Y] = [X,Y] \otimes 1 + 1 \otimes [X,Y] = \Delta_U [X,Y].$$

Mixed tensor factors commute, which is why the cross terms vanish. Coassociativity and the counit identities follow by checking the generators, since the maps are algebra homomorphisms.

The antipode identities hold on X because $-X + X = 0$. They extend to products. Suppose that they hold on u and v. Then

$$\begin{aligned} \sum S_U(u_{(1)}v_{(1)})u_{(2)}v_{(2)} &= \sum S_U(v_{(1)})S_U(u_{(1)})u_{(2)}v_{(2)} \\ &= \varepsilon_U(u) \sum S_U(v_{(1)})v_{(2)} = \varepsilon_U(u)\varepsilon_U(v)1. \end{aligned}$$

The first equality uses that S_U reverses products. The second uses the identity for u, and the third uses the identity for v. The other antipode identity is proved the same way.

The two algebras have opposite kinds of commutativity. The coefficient algebra is commutative. It is cocommutative exactly when K is abelian, because coefficients separate points. The enveloping algebra is cocommutative, as the generator formulas (28.1) show, but is generally noncommutative.

The compact real form also defines an involution on U by $X^* = -X$ for real $X \in \mathfrak{k}$, extended antilinearly with $(uv)^* = v^*u^*$. For real X, Y, applying $*$ to the defining relation gives $YX - XY + [X,Y]$, which is again zero. Thus $*$ is well defined. Applying it twice fixes the generators, so it is an involution.

In every finite-dimensional unitary representation,

$$d\Pi(u^*) = d\Pi(u)^*.$$

This follows first for real generators, whose operators are skew-adjoint, then for sums and products. The involution remembers the compact real form inside its complexification.

28.2 Differentiating coefficients

For $X \in \mathfrak{k}$, let

$$(R_X f)(g) = \left.\frac{d}{dt}\right|_0 f(g \exp(tX)).$$

This is the left-invariant vector field generated by X, acting on functions. The letter R records multiplication on the right in the argument. It is distinct from the right regular group action, although differentiating that action gives this operator. By Lecture 1, $[R_X, R_Y] = R_{[X,Y]}$. Hence, by the universal property of U, these operators extend to R_u for $u \in U$.

Definition 28.1. The *coefficient pairing* is

$$\langle u, f \rangle = (R_u f)(e), \qquad u \in U,\ f \in \mathcal{R}(K).$$

The coefficient pairing is complex bilinear. Its angle brackets denote a pairing, not a Hermitian inner product. For a coefficient, direct differentiation gives

$$R_X c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)d\Pi_V(X)v), \qquad \langle u, c^V_{\lambda,v}\rangle = \lambda(d\Pi_V(u)v). \tag{28.2}$$

The second identity follows for products of generators and then by linearity. Since the pairing was defined on functions by differential operators, it does not depend on the chosen coefficient expression.

The order matters. For two real generators $X, Y \in \mathfrak{k}$,

$$\langle XY, f\rangle = \partial_t\partial_s|_0 f(\exp(tX)\exp(sY)).$$

Indeed, for $f = c^V_{\lambda,v}$ we have $R_X R_Y f(g) = \lambda(\Pi_V(g)d\Pi_V(X)d\Pi_V(Y)v)$. Interchanging X and Y would reverse the matrix product. Subtracting the pairing of YX with f from the pairing of XY with f gives $\langle [X,Y], f\rangle$.

Proposition 28.2 (Hopf pairing identities). *Write finite coproducts as $\Delta_U u = \sum u_{(1)} \otimes u_{(2)}$ and $\Delta_{\mathcal{R}} f = \sum f_{(1)} \otimes f_{(2)}$. Then*

$$\begin{aligned}
\langle ab, f\rangle &= \sum \langle a, f_{(1)}\rangle\langle b, f_{(2)}\rangle,\\
\langle u, fh\rangle &= \sum \langle u_{(1)}, f\rangle\langle u_{(2)}, h\rangle,\\
\langle 1, f\rangle &= \varepsilon_{\mathcal{R}} f, \qquad \langle u, 1\rangle = \varepsilon_U u,\\
\langle S_U u, f\rangle &= \langle u, S_{\mathcal{R}} f\rangle.
\end{aligned}$$

Proof. Choose a basis (e_j) of V with dual (e^j). For $f = c^V_{\lambda,v}$,

$$\Delta_{\mathcal{R}} f = \sum_j c^V_{\lambda,e_j} \otimes c^V_{e^j,v}.$$

Applying the right side of the first identity gives

$$\sum_j \lambda(d\Pi(a)e_j)\, e^j(d\Pi(b)v) = \lambda(d\Pi(a)d\Pi(b)v).$$

This is $\langle ab, f\rangle$ by the coefficient formula (28.2).

For the second identity, $R_X(fh) = (R_X f)h + f(R_X h)$ is the ordinary Leibniz rule. Iterating it gives exactly the coproduct (28.1). For instance,

$$\Delta_U(XY) = XY \otimes 1 + X \otimes Y + Y \otimes X + 1 \otimes XY.$$

Thus evaluation at e proves the claim for every product and hence for every u. The identity $\langle 1, f\rangle = \varepsilon_{\mathcal{R}} f$ follows because R_1 is the identity. Applying derivatives to constants gives $\langle u, 1\rangle = \varepsilon_U u$.

Finally, write $\Pi^\vee$ for the dual representation on V^*, given by $\Pi^\vee(g)\mu = \mu \circ \Pi_V(g)^{-1}$. The function $S_{\mathcal{R}} c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)^{-1}v)$ is the coefficient of $\Pi^\vee$ with vector λ and with the functional given by evaluation at v. On V^*, a real generator acts by $d\Pi^\vee(X)\mu = -\mu \circ d\Pi(X)$. For every $u \in U$,

$$d\Pi^\vee(u)\mu = \mu \circ d\Pi(S_U u).$$

This holds for generators and extends multiplicatively because S_U reverses products. Apply the coefficient formula (28.2) to this coefficient of $\Pi^\vee$, and then to $c^V_{\lambda,v}$. Both sides of the last identity equal $\lambda(d\Pi(S_U u)v)$, which proves it. ∎

Hence the pairing makes the product of each algebra dual to the coproduct of the other. In this algebraic form, it relates the infinitesimal action to group multiplication. The proposition does not assert that one of these algebras is the entire linear dual of the other.

A noncommutative calculation

Take $K = \mathrm{SU}(2)$, so $\mathfrak{k}_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$. We test the order identity in its defining representation. Put

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These are complexified generators. The derivative is extended complex-linearly; we are not claiming that $\exp(tE)$ belongs to $\mathrm{SU}(2)$ for real t. Let $D_{ij}(g) = g_{ij}$ be coefficients of its defining representation. Then

$$EF = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad FE = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

It follows that $\langle EF, D_{11}\rangle = 1$, whereas $\langle FE, D_{11}\rangle = 0$. Their difference is $\langle H, D_{11}\rangle = 1$. The pairing therefore distinguishes EF from FE in U, even though its other argument belongs to a commutative function algebra.

The coefficient coproduct gives the same answer:

$$\Delta_{\mathcal{R}} D_{11} = D_{11} \otimes D_{11} + D_{12} \otimes D_{21}.$$

Pairing the first tensor factor with E and the second with F yields

$$E_{11}F_{11} + E_{12}F_{21} = 0 + 1 = 1.$$

Pairing the first factor with F and the second with E yields zero. Thus a commutative algebra can have a coproduct that encodes a noncommutative group law. Commutativity of the product does not by itself decide cocommutativity of the coproduct.

The regular actions on coefficients

Let $L_g f(x) = f(g^{-1}x)$ and $R_g f(x) = f(xg)$, as in the compact harmonic-analysis lectures. Here R_g denotes the group action. The symbols R_X and R_u denote differential operators. The pairings of the translates of a coefficient are

$$\langle u, L_g c_{\lambda,v}\rangle = \lambda\big(\Pi(g^{-1})d\Pi(u)v\big),$$

$$\langle u, R_g c_{\lambda,v}\rangle = \lambda\big(d\Pi(u)\Pi(g)v\big).$$

For the first formula, $L_g c$ is the coefficient with functional $\lambda \circ \Pi(g^{-1})$. For the second, $R_g c$ has vector $\Pi(g)v$. Inserting these into the coefficient formula (28.2) proves both identities without any operator-domain question.

Notice the positions of the group matrix and the derivative matrix. Left translation changes the dual variable. Right translation changes the vector variable. This is the same tensor order $V^* \otimes V$ that appeared in the Peter–Weyl theorem. The enveloping-algebra pairing continues that convention rather than introducing a second one.

28.3 What the dual contains

For any unital algebra A, its *finite dual* A° consists of the functionals that vanish on a two-sided ideal of finite codimension. Equivalently they are matrix coefficients of finite-dimensional A-modules. Here is the elementary equivalence.

A finite-dimensional module gives a homomorphism $A \to \mathrm{End}(V)$ with finite-codimensional kernel; every coefficient vanishes on that kernel. Conversely, if φ vanishes on such an ideal I, use the finite-dimensional regular module A/I, vector $1 + I$ and functional $\bar{\varphi}$. Then $\varphi(a) = \bar{\varphi}(a(1 + I))$.

By the coefficient formula (28.2), the pairing therefore maps $\mathcal{R}(K)$ into U°. Only the modules that integrate to the chosen compact group occur in this image. A finite-dimensional representation of $\mathfrak{k}_{\mathbb{C}}$ need not respect the periods of K or the restrictions imposed by a central quotient.

> **Proposition 28.3** (Injectivity for connected groups). *If K is connected, the map $\mathcal{R}(K) \to U^\circ$ from the pairing is injective.*

Proof. Suppose $\langle u, f\rangle = 0$ for every u. Since every sum of coefficients is one coefficient of a direct sum, we may write $f = c^V_{\lambda,v}$. For any finite list $X_1, \dots, X_s \in \mathfrak{k}$, expand the finite-dimensional matrix exponentials in

$$f(\exp(t_1X_1)\cdots\exp(t_sX_s)) = \lambda\big(e^{t_1 d\Pi(X_1)}\cdots e^{t_s d\Pi(X_s)}v\big).$$

These power series converge absolutely for every real or complex tuple $(t_1, \dots, t_s)$. The Taylor coefficient of $t_1^{a_1}\cdots t_s^{a_s}$ is

$$\frac{\langle X_1^{a_1}\cdots X_s^{a_s}, f\rangle}{a_1!\cdots a_s!} = 0.$$

Thus the displayed function vanishes for all real tuples. A connected Lie group is generated by its exponentials (Lecture 2), so $f = 0$ on K. ∎

For disconnected K, the same argument detects only the identity component. A coefficient may vanish on that component and be nonzero elsewhere. Thus connectedness cannot simply be removed.

Example 28.4 (The circle). Let $K = \mathrm{U}(1)$ and choose X with $\exp(tX) = e^{it}$. Then

$$U = \mathbb{C}[X], \qquad \mathcal{R}(K) = \mathbb{C}[z, z^{-1}], \qquad \langle X^k, z^n\rangle = (in)^k.$$

The image consists of finite linear combinations of the evaluation functionals $p(X) \mapsto p(in)$, $n \in \mathbb{Z}$. The finite dual is larger. Evaluation at $X = 1$ belongs to U° but is not in this image. If it were a combination supported on $n_1, \dots, n_s$, the polynomial $p(X) = \prod_j (X - in_j)$ would have zero value under that combination, but $p(1) \neq 0$.

The full linear dual is larger still. It contains the functional L given by $L(X^k) = k!$. By the equivalence above, every element of U° is a matrix coefficient of a finite-dimensional module. Suppose that X acts on that module by the operator A and that the coefficient uses the vector v and the functional λ. Then the coefficient takes the value $\lambda(A^k v)$ on X^k. Choose a norm on the module, with the corresponding operator and dual norms. This value is then bounded in absolute value by $C\|A\|^k$, where $C = \|\lambda\|\|v\|$. The sequence $k!$ has no such exponential bound. Thus L is not even in U°.

28.4 Global forms and tensor reconstruction

The groups SU(2) and SO(3) have isomorphic real Lie algebras, hence the same complex enveloping algebra after this identification. Their coefficient algebras retain different representations. Pullback through $p : \mathrm{SU}(2) \to \mathrm{SO}(3)$ gives

$$p^*\mathcal{R}(\mathrm{SO}(3)) = \{f \in \mathcal{R}(\mathrm{SU}(2)) : f(-g) = f(g)\}. \tag{28.3}$$

Indeed, a representation descends exactly when $-I$ acts trivially. On V_m it acts by $(-1)^m$. The algebraic Peter–Weyl decomposition therefore identifies the right side with precisely the coefficient blocks for even m. Odd blocks have been excluded, not annihilated by differentiation. Both coefficient algebras inject into the same finite dual, with different images.

To recover the group, keep the category of finite-dimensional continuous complex representations of K, all its intertwiners, and its forgetful functor ω. This functor sends each representation to its underlying vector space and each intertwiner to its linear map.

A *tensor automorphism* of ω is a family of maps $\theta_V \in \mathrm{GL}(V)$, one for each representation V, such that

$$\theta_W T = T\theta_V \quad \text{for every intertwiner } T : V \to W, \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

Here $\mathbb{C}$ is the trivial representation. A tensor automorphism is *unitary* if every θ_V is unitary for every invariant inner product on V. Write $\mathrm{Aut}_\otimes(\omega)$ for the group of tensor automorphisms and $\mathrm{Aut}^u_\otimes(\omega)$ for its subgroup of unitary tensor automorphisms.

Each $g \in K$ supplies the unitary tensor automorphism $\theta_V = \Pi_V(g)$. A tensor automorphism θ corresponds to the algebra homomorphism $\mathcal{R}(K) \to \mathbb{C}$ given by $c^V_{\lambda,v} \mapsto \lambda(\theta_V v)$. Unitary tensor automorphisms correspond to $*$-homomorphisms. Exercises 14.4–14.5 prove this correspondence.

Let $K_\mathbb{C}$ be the complexification of K introduced in Lecture 14. It is a complex Lie group, and its identity component is reductive. The reconstruction theorem is cited from Bröcker–tom Dieck, Sections III.7–8. It identifies

$$K \cong \mathrm{Aut}^u_\otimes(\omega), \qquad K_\mathbb{C} \cong \mathrm{Aut}_\otimes(\omega).$$

The compact group has the topology of pointwise convergence on the representation spaces. In words, every unitary tensor automorphism is $\Pi_V(g)$ for a unique $g \in K$. Every tensor automorphism is given by a unique element of $K_\mathbb{C}$, acting through the holomorphic extensions of the representations Π_V to $K_\mathbb{C}$. The unitary condition distinguishes the compact target from its complexification.

For the circle, write $\mathbb{C}_n$ for the one-dimensional representation $z \mapsto z^n$ of U(1), where $n \in \mathbb{Z}$. Every finite-dimensional representation of U(1) is a direct sum of copies of these representations. Naturality makes θ act on every copy of $\mathbb{C}_n$ by one scalar $a_n \in \mathbb{C}^\times$. The tensor rule, the isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \cong \mathbb{C}_{m+n}$ and the unit rule give $a_{m+n} = a_m a_n$ and $a_0 = 1$. Therefore $a_n = a^n$ with $a = a_1 \in \mathbb{C}^\times$. Every such choice of a defines a tensor automorphism. This tensor automorphism is unitary exactly when $|a| = 1$. This recovers $\mathbb{C}^\times$ and U(1) respectively.

For SU(2), let $V = \mathbb{C}^2$ be the defining representation. The alternating tensor $e_1 \otimes e_2 - e_2 \otimes e_1$ is invariant. Since $\theta_{V\otimes V} = \theta_V \otimes \theta_V$ and $\theta_{\mathbb{C}} = 1$, naturality with the intertwiner from $\mathbb{C}$ given by this tensor forces $\theta_V \otimes \theta_V$ to preserve it. Because $\theta_V \otimes \theta_V$ multiplies this tensor by $\det \theta_V$, we get $\det \theta_V = 1$. Every irreducible representation is a symmetric power of V, and direct sums and intertwiners determine the rest of the tensor automorphism.

Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ acts on these tensor constructions and gives a tensor automorphism, using the complexification from Lecture 14. Hence the complex target is $\mathrm{SL}(2,\mathbb{C})$. Imposing unitarity on V gives SU(2). For the even-weight category the central element $-I$ acts as the identity tensor automorphism, so reconstruction instead yields the quotient global group.

A circle quotient and its representation lattice

There is a smaller example of global information in a coefficient subalgebra. Let $\mu_N \subset \mathrm{U}(1)$ be the Nth roots of unity. The quotient is again a circle, but keep the quotient map and the identification of Lie algebras fixed. A character z^n descends through this map exactly when

$$\zeta^n = 1 \quad \text{for every } \zeta \in \mu_N, \qquad \text{or equivalently } N \mid n.$$

Thus the pulled-back coefficient algebra is

$$\mathbb{C}[z^N, z^{-N}] \subset \mathbb{C}[z, z^{-1}].$$

Under the same infinitesimal pairing, its evaluation points are $iN\mathbb{Z}$, instead of $i\mathbb{Z}$. The real Lie algebras of these two circles are isomorphic. The quotient map, however, has changed which periods a representation must respect. Rescaling an abstract Lie algebra isomorphism does not erase the specified quotient map.

The subalgebra remains closed under the Hopf operations. For its generator $w = z^N$,

$$\Delta w = w \otimes w, \qquad \varepsilon w = 1, \qquad Sw = w^{-1}, \qquad w^* = w^{-1}.$$

Reconstruction from this algebra and its unitary structure recovers the quotient circle together with its representation lattice. The same calculation with even highest weights gives the SU(2) to SO(3) distinction. In both cases, keeping a Lie algebra while changing the allowed coefficient subalgebra changes the global representation problem.

28.5 What a quantum-group course changes

The structures above give a concrete starting point for deformation. The coefficient algebra encodes the multiplication of group elements in its coproduct. The enveloping algebra encodes the infinitesimal bracket in its associative product. The pairing between them relates these two operations. A deformation can change these operations while retaining the Hopf identities.

Classical object	Product	Coproduct
$\mathcal{R}(K)$	commutative	usually not cocommutative
$U(\mathfrak{k}_{\mathbb{C}})$	usually not commutative	cocommutative

A noncommutative coefficient algebra is no longer an algebra of scalar-valued functions on an ordinary group. A noncocommutative coproduct changes how tensor products of representations carry an action. The order of tensor factors can then matter. Associativity of the tensor product of representations still follows from coassociativity.

A braiding is additional structure, not an automatic property of a Hopf algebra. In suitable quasitriangular examples on the enveloping side, an R-matrix supplies an intertwiner between the two tensor orders. The coefficient-side formulation uses the corresponding coquasitriangular structure. The future course will define these notions and prove their compatibility identities. They are not needed for any theorem here.

The $*$-structure also remains substantive. An arbitrary Hopf $*$-algebra is not automatically a compact quantum group; positivity and analytic completion require further hypotheses. Similarly, a representation category does not automatically become a fusion category. For an infinite compact Lie group there are infinitely many irreducibles. Finite fusion categories in quantum-group examples require particular constructions and assumptions.

The useful preparation is therefore the classical bookkeeping developed in this course: distinguish an algebraic coefficient space from a completion, track coproduct order, keep the real or unitary structure, and retain the global representation lattice. These are the data a deformation must replace or preserve. The theory of deformation parameters, $U_q(\mathfrak{g})$ and roots of unity begins in the later course.

Self-checks

1. For the circle, compute $\langle X^2, z^{-3} \rangle$.

Answer. It is $(-3i)^2 = -9$. The pairing is bilinear, so there is no complex conjugation in this calculation.

2. Does $U(\mathfrak{sl}(2,\mathbb{C}))$ alone decide whether V_1 (spin $1/2$) is allowed?

Answer. No. It is allowed for $\mathrm{SU}(2)$ and excluded for $\mathrm{SO}(3)$. The chosen coefficient subalgebra, or equivalently the allowed global representations, records that distinction.

3. For the circle tensor automorphism $a = 2$, what are $\theta_{\mathbb{C}_2}$ and $\theta_{\mathbb{C}_{-1}}$? Is it unitary?

Answer. They are 4 and $1/2$. It is a complex tensor automorphism, but it is not unitary and does not represent an element of the compact circle.

Summary

Differentiation pairs representative functions with the enveloping algebra. For connected K, the pairing embeds them in the finite dual. There the global group selects a specific subalgebra. Tensor reconstruction recovers K when unitary structure is kept and $K_{\mathbb{C}}$ when it is forgotten. The coefficient and enveloping Hopf structures provide the classical starting point for quantum groups.

Reading

Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Sections III.1 and III.7–8: coefficient algebras, reconstruction and complexification. The enveloping-algebra and tensor-product inputs are from the prerequisite Lie Algebras course. Lecture 14 contains the coefficient Hopf identities used here. The quantum-group paragraph is an outlook, not an additional classification theorem.